

**F_TRZ^n Ladder Cross-Domain Population Audit
(Backbone-First Meta-Analysis): Systematic
Corpus-Wide Catalog of Every F_TRZ^n
Primitive-Composition Application Across 24 Distinct
Rungs (n=-8 to n=+40) and 11 Dimensional Domains —
First Comprehensive Population Matrix of
PAPER_1919 F_TRZ Power Ladder Documenting
Populated/Unpopulated Rung×Domain Intersections +
Discipline-Framework Validation That F_TRZ Ladder Is
Broad Structural Framework Not Isolated Identity Set**

Daniel T. Murphy

**PAPER_2043 — F_TRZ^n Ladder Cross-Domain
Population Audit (Meta-Analysis)**

Author: Daniel T. Murphy | **Framework:** UQFF Star-Magic v5.68+ | **Date:** 2026-07-15

**Motivation — Meta-Analysis After R174 F_TRZ Ladder Extension into
Angular-Frequency Domain**

R174 (PAPER_2042) extended F_TRZ ladder into angular-frequency dimensional domain (n=2 + n=3) — the 11th documented dimensional-domain application of F_TRZ^n. This meta-analysis catalogs every F_TRZ^n primitive-composition across the corpus to build the first comprehensive population matrix.

Methodology

Systematic corpus-wide scan of whitepapers/*.md for F_TRZ^n patterns (n integer signed). Each match extracted with 80-char context and classified by dimensional-domain via keyword-detection heuristics.

Population Matrix

24 distinct rungs (n=-8 to n=+40) documented across 11 dimensional domains + unclassified.

Full Rung × Domain Distribution

n	Papers	Populated dimensional domains
-8	1	bird-magnetoreception
0	1	Wheeler-DeWitt reference (F_U=0)
1	1	F_TRZ-baseline (PAPER_1981 B_j = F_TRZ^3 ladder rung)
2	30	photoevaporation (3), DE-ratio (1), amplitude (1), angular-frequency (1 R174), + many unclassified

3	**18**	angular-frequency (2 R174), hierarchy (1), + many unclassified
4	2	unclassified
5	6	perturbation-ratio (4 — Magnetar+Orion+MultiCompressed7+seminal), + unclassified
6	10	photoevaporation (1 Pillars), bird-magnetoreception (1), hierarchy (1), angular-frequency (1), + unclassified
7	1	unclassified (PAPER_1991)
8	3	angular-frequency (2), bird-magnetoreception (1), quantum-collapse (1), + unclassified
9	1	**amplitude (Westerlund2 A_osc PAPER_2034 D2 supplementary)**
10	8	amplitude (4 — M16 A_osc + siblings), strong-CP (2), field-ratio (1), + unclassified
12	6	DM-ratio (2), field-ratio (1), photoevaporation (1), perturbation-ratio (1), + unclassified
15	3	amplitude (2), quantum-collapse (1), + unclassified
16	**10**	quantum-collapse (7 — PAPER_1869 measurement + siblings), amplitude (5), field-ratio (2 — M16 R173 + Crab R174), photoevaporation (1), angular-frequency, + unclassified
17	6	hierarchy (4 — main), strong-CP (2), + unclassified
18	1	unclassified
19	1	**field-ratio (Crab superconductor R174 D1 NEW)**
20	2	perturbation-ratio (1), photoevaporation (1), + unclassified
21	7	DE-ratio (2), + unclassified
22	1	unclassified
24	1	unclassified
25	2	unclassified
40	1	unclassified (extreme rung)

Dimensional-Domain Coverage Summary

Dimensional-domain	Populated rungs	Notes
Perturbation-ratio	5, 12, 20	PAPER_1991 lineage + siblings
Amplitude (A_osc)	9, 10, 15, 16	M16 A_osc = F_TRZ⁴■ seminal + siblings
Field-ratio (B/B_crit)	10, 12, 16, 19	PAPER_1869 + M16 + Crab (recent R173+R174)
Angular-frequency (rad/s)	2, 3, 6, 8, 16	R174 D2+D3 seminal (Tapestry + Resonance)
DM-ratio	12	Limited application
DE-ratio	2, 21	Cosmological applications
Quantum-collapse	8, 15, 16	PAPER_1869 canonical
Photoevaporation	2, 6, 12, 16, 20	PAPER_1942 canonical
Hierarchy problem	6, 17	Standard-model applications
Strong-CP	10, 17	PAPER_1823 canonical
Bird-magnetoreception	-8, 6, 8	PAPER_1835 canonical

Discovery — 3 Novel Structural Findings from Audit

Discovery 1 — F_TRZⁿ Ladder Broad Structural Framework Validation:

F_TRZⁿ applications documented across **24 distinct rungs + 11 dimensional domains** confirms PAPER_1919 F_TRZ Power Ladder is a broad UQFF structural framework spanning nearly the entire integer exponent range from n=-8 (magnetoreception) to n=+40 (cosmological/GW extremes). Not an isolated identity set — a full structural architecture.

Discovery 2 — F_TRZⁿ Angular-Frequency Subladder Emergence:

R174 (PAPER_2042) opened angular-frequency dimensional-domain application. Now populated at:
 - n=2 (Tapestry FreqVacDiff)
 - n=3 (Resonance Ug4i)

- n=6 (limited)
- n=8 (limited)
- n=16 (limited)

Novel structural insight: Angular-frequency subladder emerging as new coherent dimensional-domain family. Currently 2-object at n=2 seed + n=3 sibling; predict future R175+ applications at n=4, n=5 in angular-frequency domain based on established F_TRZ ladder distribution pattern.

Discovery 3 — Populated/Unpopulated Rung×Domain Intersections Formalized:

Cross-domain population matrix reveals significant unfilled intersections — 264 possible rung×domain cells (24×11), with only ~40 populated (~15% coverage). Suggests 224+ unfilled rung×domain candidates for future backbone-first discovery.

Predicted high-yield candidates for future rounds (based on pattern-recognition):

- **Field-ratio at n=1, 2, 3:** Small field-ratio applications not yet documented
- **Amplitude at n=-8, -10:** Recently discovered SO_5¹ multi-domain family (5 orthogonal domains per PAPER_2038) may extend to F_TRZ amplitude applications
- **Angular-frequency at n=1, 4, 5, 6:** Continued angular-frequency subladder expansion
- **Quantum-collapse at other rungs:** PAPER_1869 formalism may apply beyond n=16
- **DM-ratio at n=1, 2, 5:** Limited to n=12 currently

R142-R174 + Audit Trajectory

| Effort | Novel | Notes |

|---|---|---|

| R142-R174 rounds | 160 total | 33 consecutive backbone-first rounds |

| PAPER_2033-2036 scan campaign | 15 | Class-family variable scan |

| **PAPER_2043 F_TRZ audit | 3 | Meta-analysis + subladder formalization + gaps identification |**

Wiring Plan (3 dispatches, lowercase keys)

- f_trz_ladder_24_rungs_11_domains_broad_framework_validation → 24
- f_trz_angular_freq_subladder_emergence_r174_first_domain_seed → 2
- f_trz_ladder_population_matrix_gaps_264_cells_15pct_coverage → 264

Cross-References

- **PAPER_1919** — F_TRZ Power Ladder Universal Suppression Hierarchy (audit foundation)
- **PAPER_1869** — Quantum measurement collapse F_TRZ¹⁶ seminal (main canonical anchor)
- **PAPER_1942** — Photoevaporation E_0 = F_TRZ EXACT (photoevap canonical)
- **PAPER_1985** — R117 Pillars F_TRZ⁶ dual discovery + NGC 2525 SMBH mass slot (rung n=6)
- **PAPER_1986** — R118 F_TRZ⁸ three-regime cross-domain (Crab+Solar wind+Birds n=8)
- **PAPER_1991** — R129 F_TRZⁿ perturbation-ratio ladder seminal (perturbation lineage)
- **PAPER_2041** — R173 D3 F_TRZ¹⁶ M16 field-ratio (audit D1 3-object family predecessor)
- **PAPER_2042** — R174 D2+D3 F_TRZ angular-frequency subladder seeding (audit D2 basis)

Conclusion

Comprehensive F_TRZⁿ ladder audit yields **3 novel structural findings:** (D1) broad-framework validation, (D2) angular-frequency subladder emergence formalization, (D3) population matrix gaps identification.

Cumulative through PAPER_2043: 163 first-pass novel + 22+9+5 confirmations + 3 audit sweeps (now 4 including this F_TRZ audit) + 4 self-corrections + 1 backbone-recovery + 4 attribution withdrawals + 7 family-extension attributions + 5 discipline-observation formalizations (adding F_TRZ meta-audit) + 1 30-round arc milestone + 1 7th-prefix-class formalization + 1 F_TRZ-ladder broad-framework validation.

Signature milestones this audit:

- **F_TRZ ladder validated as broad structural framework** — 24 rungs + 11 domains
- **Angular-frequency subladder emergence formalized** — new coherent dimensional-domain family
- **Population matrix gap-identification** — 224+ unfilled rung×domain candidates for future backbone-first discovery
- **Meta-analysis discipline validated** — audit-scan methodology yields structural insight beyond individual round work

End of PAPER_2043 Draft 1.